Chapter 1, Problem 9PP

Problem

Referring to the residential rate schedule in Example 1.9, calculate the average cost per kWh if only 350 kWh are consumed in July when the family is on vacation most of the time.

Example 1.9:

A homeowner consumes 700 kWh in January. Determine the electricity bill for the month using the following residential rate schedule:

Base monthly charge of \$12.00.
First 100 kWh per month at 16 cents/kWh.
Next 200 kWh per month at 10 cents/kWh.
Over 300 kWh per month at 6 cents/kWh.

Solution:

We calculate the electricity bill as follows.

$$\begin{aligned} \text{Base monthly charge} &= \$12.00 \\ \text{First 100 kWh @ \$0.16/k Wh} &= \$16.00 \\ \text{Next 200 kWh @ \$0.10/k Wh} &= \$20.00 \\ \text{Remaining 400 kWh @ \$0.06/k Wh} &= \$24.00 \\ \text{Total charge} &= \$72.00 \end{aligned}$$

$$\text{Average cost} = \frac{\$72}{100 + 200 + 400} = 10.2 \text{ cents/kWh}$$

Step-by-step solution

Step 1 of 2

The following list is the residential rate schedule for the electricity bill for the month of July.

Base monthly charges of $\$12.00$

First 100 kWh per month at 16 cents/kWh

Next 200 kWh per month at 10 cents/kWh

Over 300 kWh per month at 6 cents/kWh

Total electricity consumption, 350 kWh

Step 2 of 2

Let us calculate the electricity bill as follows:

Base monthly charges $= \$12.00$

First 100 kWh @ $\$0.16/\text{kWh} = \$16.00$

Next 200 kWh @ $\$0.10/\text{kWh} = \$20.00$

Remaining 50 kWh @ $\$0.06/\text{kWh} = \$3.00$

The total charge $= \$51.00$

Calculate the average cost.

$$\begin{aligned}\text{Average cost} &= \frac{\$51}{100+200+50} \\ &= 14.571\ \text{cents/kWh}\end{aligned}$$

Therefore, the average cost is $\boxed{14.571\ \text{cents/kWh}}$.